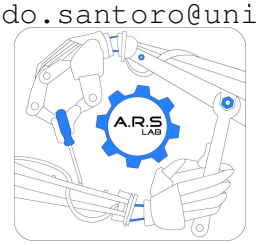
Course Intro

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Advanced Robotics and Autonomous Systems

Notes

Course Content

- **Dynamic Systems; Control Systems.** Representation of dynamic systems. State equations. Stability. Dynamic system responses. Discretization. Control Systems. PID Control. Cascading Controllers. Control Schemas.
- **System Identification.** Regressive models. Neural models.
- **Control Strategies** (alternatives to PID). Fuzzy. Linear-Quadric Optimal Control. Model Predictive Control.
- **Digital Filters.** Frequency response and Fourier analysis. Digital filter design. Predictive filters. Kalman filters.
- **Environment Mapping.** Simultaneous Location And Mapping (SLAM).
- **Middlewares for robotics.** Robot Operating System (ROS).
- **High-level programming.** Knowledge representation. Logic reasoning. Prolog. Expert Systems. Reasoning and planning. Belief-desire-intention models (BDI). BDI tools

Notes

Resources

Course slides

From `http://www.dmi.unict.it/santoro`

The Internet

Ask LLMs information about specific topics

Notes

Software Resources

- **Course software:**
  `https://github.com/UniCT-ARSLab/arslabrobotics`
- **Python Robotic Toolkit:**
  `https://github.com/petercorke/robotics-toolbox-python`
- **Godot Game Engine:**
  `https://godotengine.org/`
- **ROS2:**
  `https://www.ros.org/`
- **SWI-Prolog:**
  `https://www.swi-prolog.org/`
- **PHIDIAS BDI Tool:**
  `https://github.com/corradosantoro/PHIDIAS/`

Notes

Examination

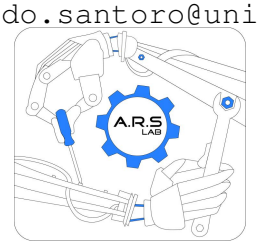
- **Project**
- **Oral Exam**

Notes

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